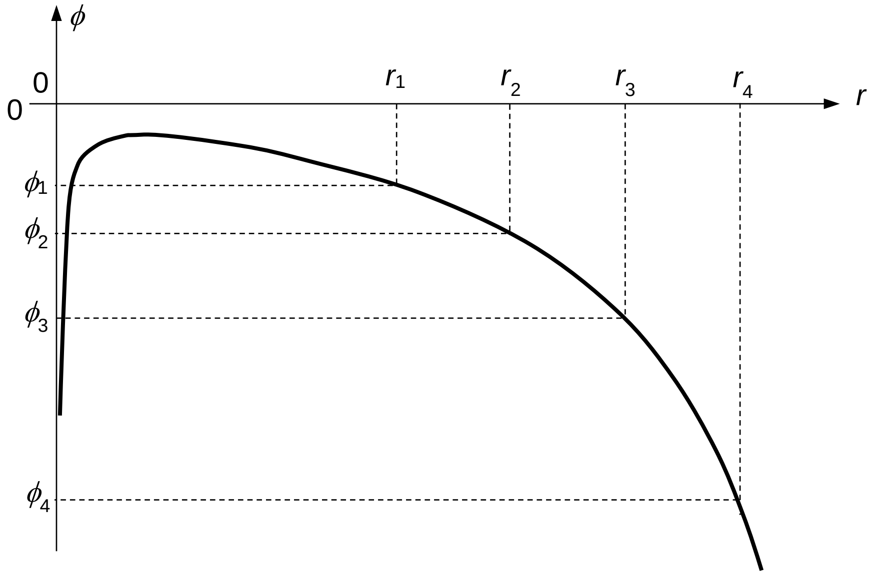
θ = (D / L) rad

14 The gravitational potential ϕ along the line joining the centres of a planet and its moon varies with the distance r from the centre of the moon as shown.



Which of the following expressions gives a value that is closest to that of the gravitational force acting on a 1 kg mass at a distance of r_2?

A $\dfrac{\phi_1 - \phi_2}{r_2 - r_1}$

B $-\dfrac{\phi_2}{r_2}$

C $\dfrac{\phi_1 - \phi_3}{r_3 - r_1}$

D $\dfrac{\phi_1 - \phi_4}{r_4 - r_1}$

Ans: C

$$F = -\frac{dE_p}{dr} = -m\frac{d\phi}{dr} \approx -1 \times \frac{\Delta\phi}{\Delta r}$$

Option A: Underestimated the gradient.

Option B: Incorrectly used $g = \frac{\phi}{r}$ .which is applicable only for a radial field. Here the *g*-field is the vector sum of the *g*-field due to the moon and the *g*-field due to the planet. The net field is not radial.

Option D: Overestimated the gradient.

15 Earth has a mass M and radius R. X is a point $5R$ from the center of the Earth.